【Opening】

大家好，今天我想介绍两篇主要论文，EmojiPrompt 和 PRISM。

Good morning, today I will talk about two main papers, EmojiPrompt and PRISM.

这两篇论文都和 prompt privacy 有关，也就是用户输入的 prompt 本身可能包含隐私信息。

Both papers are about prompt privacy, which means the user prompt itself may contain private information.

我还会简单介绍三个 baseline，分别是 Split-and-Denoise、InferDPT 和 TokEmbPriv。

I will also briefly talk about three baselines: Split-and-Denoise, InferDPT, and TokEmbPriv.

最后，我会重点说明这些论文怎么和我们现在的 soft-prompt obfuscation and recovery project 连接起来。

At the end, I will focus on how these papers connect to our soft-prompt obfuscation and recovery project.

我今天的重点不是只总结论文，而是思考它们能不能变成我们的 baseline、attack target，或者新的实验方向。

My main goal today is not only to summarize the papers, but also to think about how they can become our baselines, attack targets, or new experiment ideas.

【Slide 1: The Shared Problem】

首先，这几篇论文都有一个共同问题：cloud LLM 很强，但是用户的 prompt 本身可能就是隐私数据。

First, these papers share one problem: cloud LLMs are strong, but the user prompt itself can be private data.

比如，一个 prompt 里面可能包含用户身份、任务目标、医疗信息、金融信息，或者其他敏感文本。

For example, a prompt may include user identity, task goal, medical information, financial information, or other sensitive text.

如果我们直接把这个 prompt 发给 cloud LLM，cloud side 就可能看到完整的私有输入。

If we send this prompt directly to a cloud LLM, the cloud side may see the full private input.

所以这些方法都会在 prompt 到达 cloud 之前做某种 privacy transform。

So these methods apply some kind of privacy transform before the prompt reaches the cloud.

这个 transform 可以是 obfuscation，也可以是 perturbation，也可以是 routing。

This transform can be obfuscation, perturbation, or routing.

但是这里有一个很重要的 tradeoff。

But there is a very important tradeoff here.

一方面，变换后的 prompt 还必须让模型完成原来的任务。

On one side, the changed prompt must still let the model do the original task.

这就是 utility pressure，也就是任务性能不能掉太多。

This is the utility pressure, which means the task score should not drop too much.

另一方面，变换后的 prompt 不能泄露太多原始文本或者敏感结构。

On the other side, the changed prompt should not leak too much original text or sensitive structure.

这就是 privacy pressure，也就是攻击者不能轻易恢复用户输入。

This is the privacy pressure, which means an attacker should not easily recover the user input.

我们的角度是同时看这两件事：behavior preservation 和 recovery risk。

Our angle is to measure both things: behavior preservation and recovery risk.

也就是说，我们不只问模型还能不能做任务，也问隐藏后的 prompt 还能不能被恢复。

In other words, we do not only ask whether the model can still do the task; we also ask whether the hidden prompt can still be recovered.

【Slide 2: Why This Connects to Our Project】

这一页是我觉得最重要的连接点。

This slide is the most important connection point.

我们现在的 project 研究的是 soft-prompt obfuscation 和 prompt recovery。

Our current project studies soft-prompt obfuscation and prompt recovery.

我们把 readable system prompt 变成 learned soft prompt，然后测试它是否还能保持任务行为。

We turn a readable system prompt into a learned soft prompt, and then test whether it can still keep the task behavior.

同时，我们也测试 recovery model 能不能从 indirect evidence 里面恢复 hidden prompt structure。

At the same time, we test whether a recovery model can recover the hidden prompt structure from indirect evidence.

我们现在已经有六类 prompt taxonomy。

We already have a six-category prompt taxonomy.

这六类是 role/persona、task instruction、format specification、constraints、context knowledge 和 tool/protocol。

These six classes are role/persona, task instruction, format specification, constraints, context knowledge, and tool/protocol.

我们也有三个评价指标：Text F1、Utility F1 和 Success@0.50。

We also have three metrics: Text F1, Utility F1, and Success@0.50.

Text F1 看恢复文本和原始 prompt 的文字重合度。

Text F1 checks word overlap between the recovered prompt and the original prompt.

Utility F1 看恢复 prompt 产生的行为和原始 prompt 产生的行为是否相似。

Utility F1 checks whether the recovered prompt and the original prompt lead to similar behavior.

Success@0.50 表示如果 Utility F1 超过 0.50，我们就认为这个 prompt 被功能性恢复了。

Success@0.50 means that if Utility F1 is above 0.50, we count it as a functional recovery.

EmojiPrompt 和 PRISM 对我们有用，是因为它们都提供了新的 privacy transform。

EmojiPrompt and PRISM are useful for us because they give us new privacy transforms.

EmojiPrompt 可以作为一个 black-box generative obfuscation baseline。

EmojiPrompt can be a black-box generative obfuscation baseline.

PRISM 可以给我们一个 risk-based routing idea 和 semantic sketch idea。

PRISM can give us a risk-based routing idea and a semantic sketch idea.

所以我们的核心问题可以变成一句话。

So our main question can be one sentence.

当一个 privacy transform 还能保持任务行为时，它是否仍然泄露可以被恢复的 semantic structure?

When a privacy transform still keeps task behavior, does it also leak semantic structure that can be recovered?

这个问题正好是我们的 recovery framework 可以测量的。

This question is exactly what our recovery framework can measure.

【Slide 3: EmojiPrompt Core Idea】

接下来我先讲 EmojiPrompt。

Next, I will talk about EmojiPrompt.

EmojiPrompt 的核心想法是把 private text 变成 symbolic prompt。

The core idea of EmojiPrompt is to turn private text into a symbolic prompt.

这里的 symbolic prompt 可以包含 emojis、数学符号、逻辑符号和缩写。

The symbolic prompt can include emojis, math signs, logic signs, and short forms.

原始输入可能是 product title、review、profile 或者其他 private text。

The raw input may be a product title, a review, a profile, or other private text.

然后 LLMO，也就是 obfuscation model，把这些文本改写成 symbols。

Then LLMO, the obfuscation model, rewrites the text into symbols.

之后 LLMI，也就是 inference model，只看到 obfuscated prompt。

Then LLMI, the inference model, only sees the obfuscated prompt.

LLMI 需要基于这个 symbolic prompt 完成原来的任务。

LLMI needs to finish the original task based on this symbolic prompt.

这个方法的好处是 cloud LLM 仍然收到普通文本格式的 prompt。

The good point is that the cloud LLM still receives a normal text-form prompt.

它不需要修改模型结构，也不需要访问 hidden embedding。

It does not need to change the model structure, and it does not need hidden embeddings.

所以 EmojiPrompt 很适合 black-box cloud LLM，比如 GPT-4 这种场景。

So EmojiPrompt is useful for black-box cloud LLMs, such as GPT-4.

它的基本假设是 symbols 可以保留粗粒度语义。

Its basic idea is that symbols can keep coarse meaning.

但是 symbols 会破坏原始文字、原始词序和直接 lexical recovery。

But symbols can break the original words, the original word order, and direct word recovery.

这就是它的 privacy-utility tradeoff。

This is its privacy-utility tradeoff.

【Slide 4: EmojiPrompt Two Obfuscation Modes】

EmojiPrompt 有两种 obfuscation mode。

EmojiPrompt has two obfuscation modes.

第一种叫 reusable obfuscation。

The first one is reusable obfuscation.

它适合固定实体，比如 product title、tabular feature 或者固定 item。

It is used for fixed entities, such as product titles, table features, or fixed items.

这些实体可以先被 obfuscate 一次。

These entities can be obfuscated once first.

之后如果它们在不同 prompt 里重复出现，就直接重复使用相同的 symbolic form。

Later, if they appear again in other prompts, the same symbolic form can be reused.

这个方式的优点是便宜，而且 symbol set 比较稳定。

The good point is that it is cheap, and the symbol set stays stable.

第二种叫 non-reusable obfuscation。

The second one is non-reusable obfuscation.

它适合 free-form private text，比如 review、document 或者自然语言段落。

It is used for free-form private text, such as reviews, documents, or natural language paragraphs.

对于这种文本，EmojiPrompt 会先把文本切成 clauses。

For this kind of text, EmojiPrompt first splits the text into clauses.

然后它会打乱 clause 顺序，再分别 obfuscate 每个 clause。

Then it shuffles the clauses and obfuscates each clause one by one.

最后再把这些 symbolic clauses 组合起来。

Finally, it combines these symbolic clauses together.

这个设计会让 reconstruction 更难，因为攻击者看不到原始顺序。

This design makes reconstruction harder, because the attacker cannot see the original order.

但是它也保留了一些粗粒度语义，所以任务还是有机会成功。

But it also keeps some coarse meaning, so the task can still succeed.

对我们来说，这一点非常关键。

For us, this point is very important.

因为如果 symbols 保留了粗粒度语义，那么 recovery model 也可能利用这些 clues。

Because if symbols keep coarse meaning, a recovery model may also use these clues.

【Slide 5: EmojiPrompt Paper Figure】

原论文的 figure 展示了两件事。

The paper figures show two things.

第一，private product history 可以被转成 emoji sequences。

First, private product history can be turned into emoji sequences.

这样 cloud model 不直接看到原始 product names。

In this way, the cloud model does not directly see the original product names.

第二，obfuscator 会把文本属性映射成模糊的 symbols。

Second, the obfuscator maps text features into unclear symbols.

这些 symbols 不保留原始 surface text。

These symbols do not keep the original surface text.

但是它们可能仍然表达一些类别、情感、用途或者任务线索。

But they may still show some class, feeling, use, or task clues.

所以从 privacy 角度看，它确实隐藏了 exact words。

So from the privacy side, it does hide exact words.

但是从 recovery 角度看，它也创造了新的 human-visible semantic clues。

But from the recovery side, it also creates new human-visible semantic clues.

这就是为什么我觉得 EmojiPrompt 很适合放进我们的 evaluation。

This is why I think EmojiPrompt fits our evaluation very well.

我们的 recovery attack 可以同时使用 emoji prompt 和 behavior examples。

Our recovery attack can use both the emoji prompt and behavior examples.

然后我们可以测试 recovery model 是否能恢复 role、task、format 或 constraints。

Then we can test whether the recovery model can recover role, task, format, or constraints.

【Slide 6: EmojiPrompt Privacy-Utility Controls】

EmojiPrompt 不是简单让 LLM 随便生成 symbols。

EmojiPrompt does not simply ask an LLM to make random symbols.

它在 generative obfuscator 外面加了两个 control。

It adds two controls around the generative obfuscator.

第一个是 semantic alignment constraint。

The first one is the semantic alignment constraint.

它先用 token edit distance 来定义两个文本是否相近。

It first uses token edit distance to define whether two texts are close.

如果两个原始输入很相近，那么它们的 obfuscated outputs 也应该保持一定的 semantic alignment。

If two raw inputs are close, then their obfuscated outputs should also keep some semantic alignment.

论文用 BERTScore ratio 来操作化这个约束。

The paper uses BERTScore ratio to make this constraint work.

第二个是 LDP post-sampling constraint。

The second one is the LDP post-sampling constraint.

它不是只生成一个 obfuscated output。

It does not only generate one obfuscated output.

它会先生成多个 candidate symbolic obfuscations。

It first generates several candidate symbolic obfuscations.

然后再从一个 shared candidate set 里面 sample 最终输出。

Then it samples the final output from a shared candidate set.

这样可以调节 privacy 和 utility 之间的关系。

This can tune the relation between privacy and utility.

对我们来说，一个重要 nuance 是 EmojiPrompt 不是 learned soft prompt。

For us, one important point is that EmojiPrompt is not a learned soft prompt.

它是 text-to-symbol prompt transformation。

It is a text-to-symbol prompt transformation.

这让它更容易部署在 black-box cloud LLM 上。

This makes it easier to use with black-box cloud LLMs.

但是它也更容易被 semantic recovery attacks 探测。

But it also makes it easier to probe with semantic recovery attacks.

【Slide 7: EmojiPrompt vs Three Baselines】

EmojiPrompt 和三个 baseline 比较：SnD、InferDPT 和 TokEmbPriv。

EmojiPrompt compares with three baselines: SnD, InferDPT, and TokEmbPriv.

SnD 的方法是给 token embeddings 加 LDP noise，然后用 client denoiser 修复。

SnD adds LDP noise to token embeddings, and then uses a client denoiser to repair them.

它的优点是对 embedding privacy 有比较正式的保护。

Its strength is that it gives more formal protection for embedding privacy.

但是它需要 split model 或 embedding interface。

But it needs a split model or an embedding interface.

InferDPT 的方法是用 LDP 替换 document tokens，然后让 remote LLM 生成，再用 local model extract final output。

InferDPT replaces document tokens with LDP, asks the remote LLM to generate, and then uses a local model to extract the final output.

它的优点是适合 black-box LLM。

Its strength is that it works with black-box LLMs.

但是 perturbed text 有时候会不连贯。

But the perturbed text can sometimes be not coherent.

TokEmbPriv 主要是 BERT-style NLU 里面的 token 或 embedding privacy。

TokEmbPriv is mainly about token or embedding privacy in BERT-style NLU.

它是一个很经典的 embedding privacy baseline。

It is a classic embedding privacy baseline.

但是它不是为 open-ended prompting 设计的。

But it is not designed for open-ended prompting.

相比之下，EmojiPrompt 最大的优点是接口简单。

In comparison, the biggest strength of EmojiPrompt is its simple interface.

cloud LLM 看到的仍然是普通文本 prompt。

The cloud LLM still sees a normal text prompt.

但是这也是它的弱点。

But this is also its weakness.

因为普通文本 prompt 中的 symbols 可能泄露 semantic clues。

Because symbols in the normal text prompt may leak semantic clues.

这正好给我们的 recovery framework 提供了攻击目标。

This gives our recovery framework a good attack target.

【Slide 8: EmojiPrompt Utility Result】

EmojiPrompt 的实验结果说明它能比较好地保持 utility。

The results of EmojiPrompt show that it can keep utility quite well.

在八个真实世界数据集上，emoji obfuscation 的任务表现接近 no-obfuscation upper bound。

On eight real-world datasets, emoji obfuscation is close to the no-obfuscation upper bound.

在很多数据集上，它也超过了 SnD、InferDPT 和 TokEmbPriv 这些 baseline。

On many datasets, it also beats baselines like SnD, InferDPT, and TokEmbPriv.

这说明 symbolic obfuscation 不是只隐藏文本，它还能让模型完成任务。

This means symbolic obfuscation does not only hide text; it can also let the model do the task.

但是对我们来说，真正重要的问题不是 utility alone。

But for us, the real important question is not utility alone.

真正重要的问题是：utility 被保留的时候，privacy 是否真的足够强?

The real important question is: when utility is kept, is privacy really strong enough?

如果 symbols 能帮助 cloud LLM 完成任务，那么它们也可能帮助 attacker 恢复语义。

If symbols can help the cloud LLM do the task, they may also help an attacker recover meaning.

所以 EmojiPrompt 的好结果反而让我们的研究问题更重要。

So the good result of EmojiPrompt makes our research question even more important.

因为 utility 越高，说明 transformed prompt 里面保留的信息越多。

Because the higher the utility is, the more information the transformed prompt may keep.

我们要测的是这些信息到底会不会变成 recoverable leakage。

We need to measure whether this information becomes recoverable leakage.

【Slide 9: PRISM Core Idea】

接下来我讲 PRISM。

Next, I will talk about PRISM.

PRISM 的核心思想和 EmojiPrompt 不一样。

The core idea of PRISM is different from EmojiPrompt.

它不是把所有 prompt 都用同一种 privacy method 处理。

It does not use the same privacy method for every prompt.

它认为不同 prompt 的 privacy risk 不一样。

It says different prompts have different privacy risks.

所以系统应该根据 risk 自适应选择路径。

So the system should choose a path based on the risk.

低风险 prompt 可以直接发给 cloud。

A low-risk prompt can be sent directly to the cloud.

中等风险 prompt 可以走 cloud-edge collaboration。

A medium-risk prompt can use cloud-edge collaboration.

高风险 prompt 应该留在 edge device 上本地处理。

A high-risk prompt should stay on the edge device and be handled locally.

PRISM 先做 sensitivity profiling。

PRISM first does sensitivity profiling.

它用 NER 和 entity risk score 判断 prompt 里哪些信息敏感。

It uses NER and entity risk scores to find which parts of the prompt are sensitive.

然后 soft gating module 选择 cloud、collab 或 edge。

Then a soft gating module chooses cloud, collaboration, or edge.

这个 idea 对我们很有用。

This idea is useful for us.

因为我们也有 prompt category taxonomy。

Because we also have a prompt category taxonomy.

我们可以把 category 和 sensitivity 结合起来，设计 risk-aware obfuscation 或 routing。

We can combine category and sensitivity to design risk-aware obfuscation or routing.

【Slide 10: PRISM Routing Phase】

PRISM 的 routing phase 可以理解成三个 decision。

PRISM's routing phase can be understood as three decisions.

第一，如果 prompt 风险低，就直接 cloud-only。

First, if the prompt risk is low, use cloud-only.

这样质量最好，速度最快，但是 privacy protection 最弱。

This gives the best quality and fastest speed, but the weakest privacy protection.

第二，如果 prompt 风险中等，就用 collaborative mode。

Second, if the prompt risk is medium, use collaborative mode.

在这个模式里，cloud 不看到原始 prompt，只看到 noisy prompt 或 abstract sketch。

In this mode, the cloud does not see the raw prompt; it only sees a noisy prompt or an abstract sketch.

第三，如果 prompt 风险高，就 edge-only。

Third, if the prompt risk is high, use edge-only.

这样 privacy 最强，但是本地模型可能比较小，所以 quality 可能下降。

This gives the strongest privacy, but the local model may be small, so quality may drop.

这个设计本质上是在做 conditional privacy。

This design is basically conditional privacy.

它不是对所有 prompt 都加同样多的 noise。

It does not add the same amount of noise to every prompt.

它是根据风险决定应该花多少 privacy budget。

It decides how much privacy budget to spend based on risk.

这点对我们的 defense design 很重要。

This point is important for our defense design.

【Slide 11: PRISM Collaborative Mode】

PRISM 最有意思的是 collaborative mode。

The most interesting part of PRISM is the collaborative mode.

这个模式用在 medium-risk prompts 上。

This mode is used for medium-risk prompts.

在这个模式里，原始 prompt P 一直留在 edge device。

In this mode, the original prompt P always stays on the edge device.

系统会对 prompt 里的 sensitive entities 做 two-layer entity LDP。

The system applies two-layer entity LDP to sensitive entities in the prompt.

这样得到一个 noisy prompt P star。

This gives a noisy prompt P star.

cloud LLM 只看到 P star，而不是原始 P。

The cloud LLM only sees P star, not the raw P.

然后 cloud LLM 不生成最终答案。

Then the cloud LLM does not write the final answer.

它只生成一个 abstract semantic sketch。

It only makes an abstract semantic sketch.

这个 sketch 可以是 high-level reasoning structure、plan，或者 abstract answer。

This sketch can be a high-level reasoning structure, a plan, or an abstract answer.

最后 edge-side small language model 同时使用原始 P 和 cloud sketch 生成最终答案。

Finally, the edge-side small language model uses both the raw P and the cloud sketch to make the final answer.

这个设计的好处是 cloud 提供推理能力，但是不知道真实私有信息。

The good point is that the cloud gives reasoning power, but it does not know the real private information.

edge 拿着原始私有信息，所以可以把最终答案变得具体和准确。

The edge has the raw private information, so it can make the final answer specific and correct.

用一句话说，cloud power 被限制在 abstract reasoning 上。

In one sentence, cloud power is limited to abstract reasoning.

这个设计对我们很重要，因为 semantic sketch 本身可能成为 recovery evidence。

This design is important for us because the semantic sketch itself may become recovery evidence.

如果 sketch 包含了足够多的 role、task、format 或 tool protocol，那 attacker 可能恢复 hidden structure。

If the sketch contains enough role, task, format, or tool protocol, an attacker may recover the hidden structure.

所以 PRISM 既可以作为 defense idea，也可以作为 attack evaluation target。

So PRISM can be both a defense idea and an attack evaluation target.

【Slide 12: PRISM Adaptive Two-Layer LDP】

PRISM 的 two-layer LDP 也很值得注意。

PRISM's two-layer LDP is also important.

普通 LDP 可能只改 entity value。

Normal LDP may only change the entity value.

比如把 New York 改成 Bibury Town。

For example, it may change New York to Bibury Town.

但是 PRISM 认为只有 value-level perturbation 不够。

But PRISM says value-level perturbation is not enough.

因为 entity category 本身也可能泄露隐私。

Because the entity category itself may also leak privacy.

比如一个 prompt 里出现 diagnosis category，即使具体疾病名字被隐藏，也可能说明这是医疗问题。

For example, if a prompt has a diagnosis category, even if the disease name is hidden, it may still show that this is a medical issue.

所以 PRISM 先扰动 category，再扰动 value。

So PRISM first perturbs the category, and then perturbs the value.

它还会根据 sensitivity weight 分配 privacy budget。

It also assigns the privacy budget based on sensitivity weight.

高风险 entity 会得到更强保护。

High-risk entities get stronger protection.

低风险 entity 可以保留更多语义，让任务更容易完成。

Low-risk entities can keep more meaning, so the task is easier to finish.

这个 idea 可以启发我们。

This idea can guide us.

我们的 prompt taxonomy 也可以看成不同的 semantic dimensions。

Our prompt taxonomy can also be seen as different semantic dimensions.

比如 role/persona 可能容易恢复，tool/protocol 可能难恢复，但也可能更敏感。

For example, role/persona may be easy to recover, while tool/protocol may be hard to recover but more sensitive.

所以未来我们也可以做 category-aware obfuscation。

So in the future, we can also try category-aware obfuscation.

不同 prompt category 可以使用不同的 obfuscation strength。

Different prompt categories can use different obfuscation strength.

【Slide 13: PRISM Efficiency and Quality Result】

PRISM 的实验结果说明 adaptive routing 有明显优势。

PRISM's results show that adaptive routing has clear advantages.

Cloud-only 最快，质量最高，但是没有 privacy protection。

Cloud-only is the fastest and has the best quality, but it has no privacy protection.

Edge-only privacy 最强，但是速度慢，质量也比较低。

Edge-only has the strongest privacy, but it is slow and has lower quality.

Uniform LDP 和 Selective LDP 有 privacy protection，但是 latency 和 energy cost 很高。

Uniform LDP and Selective LDP have privacy protection, but their latency and energy cost are high.

PRISM 在 privacy-preserving methods 里面取得了最好的 tradeoff。

PRISM gets the best tradeoff among privacy-preserving methods.

它比 Uniform LDP 和 Selective LDP 更快，能耗更低，质量也更高。

It is faster, uses less energy, and has higher quality than Uniform LDP and Selective LDP.

这个结果告诉我们一个方法论 lesson。

This result gives us a method lesson.

privacy 不应该是 uniform 的。

Privacy should not be uniform.

也就是说，我们不应该对每个 prompt、每个 token、每个 category 都做一样强的保护。

That means we should not protect every prompt, every token, and every category in the same way.

更好的方法是根据 risk 来决定保护强度。

A better way is to decide protection strength based on risk.

这个 lesson 可以直接用于我们的 project。

This lesson can be used directly in our project.

我们可以根据六类 prompt 的 recovery risk 和 utility need 来设计不同的 defense。

We can design different defenses based on the recovery risk and utility need of the six prompt categories.

【Slide 14: Baseline Details】

这里我简单总结三个 baseline。

Here I will briefly summarize the three baselines.

第一个是 Split-and-Denoise，也就是 SnD。

The first one is Split-and-Denoise, or SnD.

SnD 把模型从 embedding layer 拆开。

SnD splits the model at the embedding layer.

client 先得到 token embeddings，然后加入 d-chi 或 LDP noise。

The client first gets token embeddings, and then adds d-chi or LDP noise.

server 在 noisy embeddings 上继续计算。

The server keeps computing on noisy embeddings.

最后 client 用 denoiser 恢复 useful representation。

Finally, the client uses a denoiser to recover useful representation.

第二个是 InferDPT。

The second one is InferDPT.

InferDPT 对 document tokens 做 LDP perturbation。

InferDPT applies LDP perturbation to document tokens.

它的 RANTEXT 会从 random adjacency lists 里采样替换 token。

Its RANTEXT samples replacement tokens from random adjacency lists.

remote LLM 基于 perturbed prompt 生成输出。

The remote LLM generates output from the perturbed prompt.

local model 再从输出里面 extract coherent final answer。

Then the local model extracts a coherent final answer from the output.

第三个是 TokEmbPriv 或 Privacy-Preserving BERT。

The third one is TokEmbPriv, or Privacy-Preserving BERT.

它主要保护 BERT-style NLU 里面的 token 或 embedding。

It mainly protects tokens or embeddings in BERT-style NLU.

它也说明 privacy-adaptive pretraining 可以帮助模型适应 privatized inputs。

It also shows that privacy-adaptive pretraining can help the model use privatized inputs.

这三个 baseline 的共同点是它们主要关注 token-level 或 embedding-level privacy。

The common point of these baselines is that they mainly focus on token-level or embedding-level privacy.

但是我们的 project 更关注 high-level prompt structure recovery。

But our project focuses more on high-level prompt structure recovery.

这就是我们和这些 baseline 的区别。

This is the difference between our work and these baselines.

【Slide 15: How to Combine Big Picture】

接下来是我觉得最重要的部分：怎么把这些论文和我们的 project 结合起来。

Next is the most important part: how to combine these papers with our project.

我们的现有 pipeline 是：prompt、obfuscation、model behavior、recovery attack、evaluation。

Our current pipeline is: prompt, obfuscation, model behavior, recovery attack, and evaluation.

EmojiPrompt 和 PRISM 都可以插入这个 pipeline。

EmojiPrompt and PRISM can both be added into this pipeline.

对于 EmojiPrompt，我们可以把它当作一个新的 obfuscation method。

For EmojiPrompt, we can use it as a new obfuscation method.

它不是 soft prompt，而是 text-to-symbol prompt。

It is not a soft prompt; it is a text-to-symbol prompt.

所以它可以和我们的 soft-prompt obfuscation 做 comparison。

So it can be compared with our soft-prompt obfuscation.

对于 PRISM，我们可以把它当作一个 routing layer。

For PRISM, we can use it as a routing layer.

它可以根据 prompt category 和 sensitivity 决定 prompt 应该 normal send、obfuscate send，还是 local only。

It can decide whether the prompt should be sent normally, sent after obfuscation, or kept local, based on category and sensitivity.

同时，PRISM 的 semantic sketch 也可以成为一种新的 recovery evidence。

At the same time, PRISM's semantic sketch can become a new kind of recovery evidence.

也就是说，我们不仅攻击 obfuscated prompt，也可以攻击 cloud sketch。

That means we can attack not only the obfuscated prompt, but also the cloud sketch.

如果 sketch 能帮助 edge model 生成好答案，它也可能包含 recoverable structure。

If the sketch can help the edge model make a good answer, it may also contain recoverable structure.

这个思路可以把我们的 project 从 soft prompt recovery 扩展到 general privacy transform evaluation。

This idea can extend our project from soft prompt recovery to general privacy transform evaluation.

【Slide 16: Combine EmojiPrompt with Our Project】

对于 EmojiPrompt，我建议两个方向。

For EmojiPrompt, I suggest two directions.

第一个方向是 recovery risk evaluation。

The first direction is recovery risk evaluation.

我们可以用自己的 six-category recovery pipeline 去攻击 EmojiPrompt。

We can use our six-category recovery pipeline to attack EmojiPrompt.

输入 evidence 可以包括 emoji prompt 和 behavior examples。

The input evidence can include the emoji prompt and behavior examples.

然后 recovery model 输出它猜测的 hidden prompt structure。

Then the recovery model outputs its guess of the hidden prompt structure.

我们继续使用 Text F1、Utility F1 和 Success@0.50。

We can still use Text F1, Utility F1, and Success@0.50.

这样我们可以直接比较 EmojiPrompt 和 soft prompt 哪个更容易被恢复。

In this way, we can directly compare whether EmojiPrompt or soft prompt is easier to recover.

关键问题是：emoji symbols 会不会比 soft embeddings 泄露更多语义？

The key question is: do emoji symbols leak more meaning than soft embeddings?

我猜可能会，因为 emoji prompt 是 human-visible 的。

My guess is yes, because emoji prompts are human-visible.

但是我们需要实验来验证。

But we need experiments to test this.

第二个方向是 hybrid defense。

The second direction is hybrid defense.

我们可以把 text prompt 先通过 EmojiPrompt，再转成 soft prompt。

We can first pass the text prompt through EmojiPrompt, and then turn it into a soft prompt.

这个流程是 text prompt to EmojiPrompt to soft-prompt obfuscation。

This flow is text prompt to EmojiPrompt to soft-prompt obfuscation.

也可以反过来，把 soft prompt projection 转成 emoji-like semantic sketch。

We can also do the opposite and turn soft prompt projection into an emoji-like semantic sketch.

然后我们测 utility 是否更好，leakage 是否更低。

Then we test whether utility is better and leakage is lower.

这里最重要的是不要只看 task score。

The most important thing here is not to only check the task score.

我们也要看 recovered prompt 是否仍然有 functional utility。

We also need to check whether the recovered prompt still has functional utility.

如果 recovery prompt 文字不一样，但是行为很接近，那仍然是 privacy risk。

If the recovered prompt uses different words but has similar behavior, it is still a privacy risk.

这就是 Utility F1 的价值。

This is the value of Utility F1.

【Slide 17: Combine PRISM with Our Project】

对于 PRISM，我建议从 defense 和 attack 两边结合。

For PRISM, I suggest combining it from both the defense side and the attack side.

在 defense side，我们可以借鉴它的 risk-aware routing。

On the defense side, we can use its risk-aware routing.

我们先判断 prompt 的 category 和 sensitivity。

We first judge the prompt category and sensitivity.

低风险 prompt 可以直接发送，因为 privacy risk 小，而且 utility 最好。

Low-risk prompts can be sent directly, because privacy risk is small and utility is best.

中等风险 prompt 可以用 EmojiPrompt 或 soft prompt obfuscation。

Medium-risk prompts can use EmojiPrompt or soft prompt obfuscation.

高风险 prompt 可以 local only，或者使用最强 obfuscation。

High-risk prompts can be local only, or use the strongest obfuscation.

这样我们的六类 taxonomy 就不只是分析工具，而可以变成 routing policy。

In this way, our six-category taxonomy is not only an analysis tool, but also a routing policy.

在 attack side，我们可以使用 PRISM-style semantic sketch 作为 evidence。

On the attack side, we can use PRISM-style semantic sketch as evidence.

问题是：cloud sketch 会不会泄露足够多的 role、task、format 或 tool protocol?

The question is: does the cloud sketch leak enough role, task, format, or tool protocol?

如果会，那么即使原始 prompt 没有发给 cloud，cloud side 仍然可能学到 prompt structure。

If yes, then even if the raw prompt is not sent to the cloud, the cloud side may still learn the prompt structure.

这对 PRISM 这样的 defense 是一个很重要的安全问题。

This is an important security question for a defense like PRISM.

一个具体实验可以这样做。

A concrete experiment can be done like this.

对每个 prompt category，我们先生成 original prompt。

For each prompt category, we first make the original prompt.

然后做 sensitivity profile。

Then we make a sensitivity profile.

接着选择 route 或 obfuscation method。

Next, we choose the route or obfuscation method.

再让 cloud 生成 semantic sketch。

Then we let the cloud make a semantic sketch.

最后把 sketch 和 behavior examples 交给 recovery model。

Finally, we give the sketch and behavior examples to the recovery model.

我们比较不同 route mode 下的 recovery rate。

We compare recovery rates under different route modes.

如果 collab mode 的 utility 很好，但 recovery rate 也很高，那说明 sketch 可能泄露了太多信息。

If collab mode has good utility but also high recovery rate, then the sketch may leak too much information.

【Slide 18: Proposed Research Plan】

最后，我提出一个比较具体的 research plan。

Finally, I propose a concrete research plan.

第一阶段是 reproduce。

The first stage is reproduce.

问题是：我们能不能在自己的 tasks 上实现 EmojiPrompt?

The question is: can we implement EmojiPrompt on our own tasks?

deliverable 是一个 utility table，比较 emoji、soft prompt 和 baseline methods。

The deliverable is a utility table comparing emoji, soft prompt, and baseline methods.

第二阶段是 recover。

The second stage is recover.

问题是：我们的 recovery model 能不能 infer emoji-obfuscated prompts?

The question is: can our recovery model infer emoji-obfuscated prompts?

deliverable 是 Text F1、Utility F1 和 Success@0.50。

The deliverable is Text F1, Utility F1, and Success@0.50.

第三阶段是 route。

The third stage is route.

问题是：PRISM-style routing 能不能降低 leakage，同时保持 utility?

The question is: can PRISM-style routing reduce leakage while keeping utility?

deliverable 是 risk-aware routing ablation。

The deliverable is a risk-aware routing ablation.

第四阶段是 probe。

The fourth stage is probe.

问题是：哪些 probes 最容易暴露 semantic leakage?

The question is: which probes expose semantic leakage most easily?

deliverable 是 category-specific task-matched probe set。

The deliverable is a category-specific task-matched probe set.

我觉得这个 plan 和老师说的 proactively lead the project 是一致的。

I think this plan matches the teacher's request to lead the project proactively.

因为它不是只复现别人的方法。

Because it does not only reproduce other methods.

它把这些方法变成我们的实验对象，并且继续服务于我们的核心问题。

It turns these methods into our experiment targets, and still serves our main research question.

【Closing】

总结一下，EmojiPrompt 给我们一个很好的 black-box symbolic obfuscation baseline。

To conclude, EmojiPrompt gives us a good black-box symbolic obfuscation baseline.

PRISM 给我们一个 risk-aware routing 和 semantic sketch collaboration 的思路。

PRISM gives us an idea of risk-aware routing and semantic sketch collaboration.

三个 baseline 说明已有方法主要关注 token-level 或 embedding-level privacy。

The three baselines show that existing methods mainly focus on token-level or embedding-level privacy.

但是我们的 project 可以进一步研究 high-level prompt structure leakage。

But our project can further study high-level prompt structure leakage.

所以我认为最强的 project direction 不是简单做更好的 obfuscation。

So I think the strongest project direction is not just to make better obfuscation.

更重要的是评价一种 privacy transform 在保持 behavior 的同时，是否仍然泄露 recoverable semantic structure。

More importantly, we should test whether a privacy transform still leaks recoverable semantic structure while keeping behavior.

这也是我们的 Text F1、Utility F1 和 Success@0.50 最有价值的地方。

This is also where our Text F1, Utility F1, and Success@0.50 are most useful.

我的建议是下一步先实现 EmojiPrompt baseline，然后用我们的 recovery pipeline 测它的 leakage。

My suggestion is to first implement the EmojiPrompt baseline, and then use our recovery pipeline to test its leakage.

之后再加入 PRISM-style routing，测试不同 risk level 下的 privacy-utility tradeoff。

After that, we can add PRISM-style routing and test the privacy-utility tradeoff under different risk levels.

谢谢大家，我的分享结束。

Thank you, this is the end of my talk.